Lecture 1 — MATH 327

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Contents

1. Axioms for a group

To define a group, we begin with a set G equipped with a *composition law*

$$\cdot\colon G \times G \longrightarrow G, \qquad (g,h) \longmapsto g \cdot h, \tag{1.1}$$

and a distinguished element $e \in G$, called the *identity*. Thus the basic data are written $(G, \cdot, e)$. We often suppress the dot and write gh in place of $g \cdot h$.

Definition 1.1 (Group)**.** The data $(G, \cdot, e)$ form a *group* if the following three conditions hold.

(1) Associativity. For all $g, h, k \in G$,

$$g \cdot (h \cdot k) = (g \cdot h) \cdot k. \tag{1.2}$$

(2) Identity. For every $g \in G$,

$$e \cdot g = g \cdot e = g. \tag{1.3}$$

(3) Inverses. For every $g \in G$, there exists $h \in G$ such that

$$h \cdot g = e \qquad \text{and} \qquad g \cdot h = e. \tag{1.4}$$

The first equality says that h is a *left inverse* of g; the second says that it is a *right inverse*. Thus an inverse is a two-sided inverse.

Associativity does *not* say that

$$g \cdot h = h \cdot g. \tag{1.5}$$

An operation satisfying this latter identity for all elements is called *commutative*.

Definition 1.2 (Abelian group)**.** A group whose operation is commutative is called *abelian*.

It is also useful to observe the following simple result.

Lemma 1.1 (Uniqueness of inverses)**.** *The element h in axiom **(3)** is unique.*

Proof. Suppose h_1 and h_2 both satisfy axiom **(3)** for g. Then

$$h_1 g = h_2 g = e. \tag{1.6}$$

Multiplying on the right by h_2 and using associativity gives

$$(h_1 g)h_2 = (h_2 g)h_2, \qquad \text{hence} \qquad h_1(gh_2) = h_2(gh_2). \tag{1.7}$$

Since $gh_2 = e$, it follows that $h_1 = h_2$. We write the unique inverse of g as g^{-1}. $\square$

2. Associativity, cyclic groups, and order

The associativity axiom allows us to write both

$$g \cdot (h \cdot k) \qquad \text{and} \qquad (g \cdot h) \cdot k \tag{2.1}$$

simply as

$$g \cdot h \cdot k. \tag{2.2}$$

More generally, given $g_1, \ldots, g_n \in G$, repeated application of associativity allows us to define the element

$$g_1 g_2 \cdots g_n \tag{2.3}$$

independently of how parentheses are inserted. Note again that we cannot change the order of the factors in the product unless the group is abelian.

For $x \in G$ and a positive integer n, we also write

$$x^n = \underbrace{x \cdot x \cdots x}_{n \text{ factors}}. \tag{2.4}$$

If $n = 0$, we set

$$x^0 = e, \tag{2.5}$$

where e is the identity of the group. If $n > 0$ and x^{-1} denotes the unique inverse of x, we set

$$x^{-n} = \underbrace{x^{-1} \cdot x^{-1} \cdots x^{-1}}_{n \text{ factors}}. \tag{2.6}$$

It then follows that

$$x^{-n} = (x^n)^{-1}. \tag{2.7}$$

Indeed,

$$(x^n)(x^{-n}) = \underbrace{x \cdot x \cdots x}_{n \text{ factors}} \underbrace{x^{-1} \cdot x^{-1} \cdots x^{-1}}_{n \text{ factors}} = e. \tag{2.8}$$

Definition 2.1 (Order of an element)**.** Let $x \in G$. The *order* of x, denoted $\mathrm{ord}_G(x)$ or $\mathrm{ord}(x)$, is the smallest positive integer n such that $x^n = e$. If no such n exists, we write $\mathrm{ord}(x) = \infty$ and say that x has infinite order. In particular, $\mathrm{ord}(e) = 1$.

Remark. Confusingly, we also say that if G is finite as a set then G has finite order. If G is infinite, we write $|G| = \infty$ and say that G has infinite order.

Here is a simple result.

Lemma 2.1. *Let $(G, \cdot, e)$ be a group and let $x \in G$. Write*

$$\langle x \rangle := \{x^n \mid n \in \mathbb{Z}\}. \tag{2.9}$$

Then $\langle x \rangle$ is a subgroup of G, called the cyclic subgroup generated by x.

Proof. For all $m, n \in \mathbb{Z}$, associativity and the definitions of positive and negative powers give

$$x^n x^m = x^{n+m}, \tag{2.10}$$

so $\langle x \rangle$ is closed under the group operation. Associativity is inherited from G, the identity $x^0 = e$ belongs to $\langle x \rangle$, and (2.7) shows that the inverse of x^n is x^{-n}. Thus $\langle x \rangle$ is a subgroup of G. $\square$

Here is a question to think about: certainly, if G has finite order, every element has finite order. But is the converse true? That is, if every element of G has finite order, must G be finite? Before we answer this question, let us turn to examples.

3. Examples

Example 3.1 (The integers under addition)**.** The triple

$$(\mathbb{Z}, +, 0) \tag{3.1}$$

is a group. Its composition law is

$$\mathbb{Z} \times \mathbb{Z} \longrightarrow \mathbb{Z}, \qquad (m, n) \longmapsto m + n. \tag{3.2}$$

The identity element is 0, and the inverse of m is $-m$.

Example 3.2 (The integers under multiplication: a non-example)**.** The triple

$$(\mathbb{Z}, \times, 1) \tag{3.3}$$

is *not* a group, although it is a semigroup. Multiplication is associative and 1 is an identity, so axioms **(1)** and **(2)** hold. Axiom **(3)** fails because most integers do not have multiplicative inverses in $\mathbb{Z}$.

Example 3.3 (Integers modulo n)**.** Let n be a positive integer. The set of residue classes modulo n is

$$\mathbb{Z}/n\mathbb{Z} = \{\overline{0}, \overline{1}, \overline{2}, \ldots, \overline{n-1}\}. \tag{3.4}$$

For example,

$$\begin{aligned} \overline{0} &= \{0, \pm n, \pm 2n, \pm 3n, \ldots\}, \\ \overline{1} &= \{1, 1 \pm n, 1 \pm 2n, 1 \pm 3n, \ldots\}. \end{aligned}$$

Every integer can be written $m = an + r$ with $0 \leq r < n$, and then $\overline{m} = \overline{r}$. Addition is defined by

$$\overline{m} + \overline{\ell} = \overline{m + \ell}. \tag{3.5}$$

Thus

$$(\mathbb{Z}/n\mathbb{Z}, +, \overline{0}) \tag{3.6}$$

is a group, called the *cyclic group of order n*.

In the preceding examples that are groups, the group operation is both associative and commutative.

Example 3.4 (The general linear group)**.** The set

$$\mathrm{GL}_n(\mathbb{R}) = \{A : A \text{ is an invertible } n \times n \text{ matrix with entries in } \mathbb{R}\} \tag{3.7}$$

is a group under matrix multiplication. Its identity is the identity matrix

$$I_n = \begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix}. \tag{3.8}$$

Thus the group is written $(\mathrm{GL}_n(\mathbb{R}), \cdot, I_n)$, and the composition law sends (A, B) to AB.

Example 3.5 (The special linear group)**.** Consider the following subset of the previous example:

$$\mathrm{SL}_n(\mathbb{R}) = \{A \in M_n(\mathbb{R}) : \det(A) = 1\}. \tag{3.9}$$

It is a group under matrix multiplication. Indeed, $I_n \in \mathrm{SL}_n(\mathbb{R})$, and if $A, B \in \mathrm{SL}_n(\mathbb{R})$, then

$$\det(AB) = \det(A)\det(B) = 1, \tag{3.10}$$

so $AB \in \mathrm{SL}_n(\mathbb{R})$. Moreover,

$$\det(A^{-1}) = \det(A)^{-1} = 1, \tag{3.11}$$

so $A^{-1} \in \mathrm{SL}_n(\mathbb{R})$. Associativity is inherited from matrix multiplication. Thus $(\mathrm{SL}_n(\mathbb{R}), \cdot, I_n)$ is a group.

4. Subgroups

Definition 4.1 (Subgroup)**.** Let $(G, \cdot, e)$ be a group. A subset $H \subseteq G$ is called a *subgroup* of G if H, under the restriction of the operation on G, is itself a group.

Not all subsets are groups. For example, the subset $\{0, 3\} \subset \mathbb{Z}$ is not a group under addition, since it is not closed under addition.

Proposition 4.1. *Let $(G, \cdot, e)$ be a group, and let $H \subseteq G$ be a subset satisfying the following conditions:*

(1) $e \in H$.
(2) H is closed under the group operation: if $h_1, h_2 \in H$, then

$$h_1 \cdot h_2 \in H. \tag{4.1}$$

(3) H is closed under inverses: if $h \in H$, then

$$h^{-1} \in H. \tag{4.2}$$

Then H is a subgroup of G.

Proof. The operation on H is the restriction of the operation on G. By condition (2), this operation is closed on H, and associativity is inherited from G. By condition (1), the identity e belongs to H, and it remains an identity for the restricted operation. Finally, condition (3) supplies an inverse in H for every element of H. Thus H satisfies all the group axioms. □

Remark. Suppose $H \subseteq G$ is finite. If $e \in H$ and H is closed under the group operation, then H is automatically closed under inverses. To see this, fix $h \in H$ and consider the map

$$L_h \colon H \longrightarrow H, \qquad x \longmapsto hx. \tag{4.3}$$

This map is well-defined because H is closed under the group operation. It is injective: if $hx = hy$, then multiplying on the left by h^{-1} in G gives $x = y$. Since H is finite, L_h is therefore surjective. Because $e \in H$, there exists $y \in H$ such that $hy = e$. By uniqueness of inverses in G, we have $y = h^{-1}$. Hence $h^{-1} \in H$.

Example 4.1 (The permutation group)**.** The permutation group is

$$S_n = \{\text{all permutations of } \{1, 2, \ldots, n\}\}. \tag{4.4}$$

Its group operation is composition of permutations. This is the example we will analyze in more detail next class.